



Determining the Effect of Training on Uncertainty Visualization Evaluations

Stephen M. Fiore^{1,2(✉)}, Jihye Song^{1,2}, Olivia B. Newton^{1,2},
Corey Pittman¹, Samantha F. Warta^{1,2}, and Joseph J. LaViola¹

¹ University of Central Florida Orlando, Orlando, FL, USA
cpittman@knights.ucf.edu, jjl@cs.ucf.edu

² Institute for Simulation and Training Orlando, Orlando, FL, USA
{sfiore, csong, onewton, swarta}@ist.ucf.edu

Abstract. Traditional studies in uncertainty visualization often require naive participants to complete complex, domain-specific tasks in order to examine how effectively a visualization conveys uncertainty to support decision making. However, without assessing whether participants understand such tasks, it can be difficult to determine whether differences in performance are due to a given visualization or to varying degrees of comprehension. Although training is commonly administered to non-experts, to date, training has not been a focal point in uncertainty visualization research. In this paper, we evaluated how variations in training, coupled with assessments of knowledge acquisition and application, can inform uncertainty visualization research. Overall, we found significant performance differences based on training condition, illustrating how training influences task comprehension, which in turn influences decision making. This study serves to highlight training as a critical component of uncertainty visualization studies by quantifying performance variations due to training.

Keywords: Human factors · Visualization studies · Training and evaluation

1 Introduction

Decision making under uncertain conditions continues to be a pressing challenge for operators in complex environments. There has been significant interest in visualizing complex information [1–4] and examining uncertainty visualizations to support decision making [5–7]. However, studies in this area typically focus on *knowledge application*—that is, how a visualization alters performance, as opposed to how *knowledge acquisition* alters knowledge application. When assessing the efficacy of visualizations as decision aids, it is important to ensure performance differences can be attributed to the visualization of interest, and not to varying degrees of task understanding. This is particularly important for complex decision-making tasks requiring domain-specific knowledge that may be especially challenging for non-experts, necessitating greater care to ensure such participants understand the task at hand. In this paper, we seek to demonstrate how significant performance differences in uncertainty visualization studies may be attributed to training. We first review research related to this issue. We then outline the rationale and forms of training designed for this study,

as well as the assessments developed to evaluate knowledge acquisition and application. Additionally, we introduce a unique combinatory metric, *cognitive efficiency*, devised to more finely diagnose how training alters workload and performance.

1.1 Related Work

A number of challenges have been raised about how to adequately evaluate visualizations to improve their validity, with some calling for a critical examination of approaches with more rigorous experiments [8, 9]. We identify training as a key component of such evaluations, suggesting it can strengthen uncertainty visualization research. Prior work has examined the benefit of training for decision-making tasks using uncertainty visualizations [10]. Currently, however, while it is common practice to administer some form of introductory task or training prior to experimental tasks, there is significant variability across uncertainty visualization studies. First, some studies may provide relatively little training. Second, the format of training may be inconsistent across studies. Third, assessments may not be administered to determine participants' understanding of the training. Finally, because some publications only briefly mention training, it may be difficult to determine its form and how comprehensive it is [11]. Yet, training may have a significant effect on outcomes in visualization studies. For instance, the way participants are trained may alter how they use visualizations during tasks [12], suggesting that inconsistent training, even for the same task, may lead to performance differences. In a similar vein, lack of training has been explicitly listed as a potential cause of performance variation in tasks involving uncertainty visualizations [13]. One factor influencing performance is mental effort, or workload [14]. Effective uncertainty visualizations, combined with training, have the potential to improve decision making by externalizing information and reducing workload, and can lead to non-expert performance similar to that of experts [15]. However, without adequate training, introducing unfamiliar visualization systems to complex decision-making tasks may increase workload, even in experts [16]. Additionally, compared to experts, non-experts may be more susceptible to impaired performance and workload during decision-making tasks involving uncertainty, particularly when only limited training is provided [17].

In light of these issues, we propose training should be given closer attention as a factor influencing performance in uncertainty visualization studies. To investigate the effect of training on performance in this context, we developed a scenario designed for non-experts to assess performance, workload, and cognitive efficiency.

1.2 Scenario

One of our goals was to develop a realistic scenario that would lower the barrier of entry for a comparatively more complex scenario. To this end, we created a simulated civilian scenario to serve as an analogue to a Naval Intelligence Unit operational task—specifically, drug interdiction. Tasks of this nature require the integration of multiple components into a single decision outcome, sometimes referred to as Course of Action (COA) selection task [18]. Relevant decision processes in our scenario were meant to mimic those found in an environment where operators are faced with variable forms of

uncertainty. Instead of intercepting drugs, our scenario requires participants to obtain supplies lost at sea in order to host a party on a tropical island. The COA selection task takes the form of a game that requires participants to retrieve a variety of supplies before their rivals steal them by sending vehicles and crew members to locations where they might be. Participants are instructed to maximize the amount and variety of assets (party supplies, alcohol, and food) while minimizing resource use (boats and crew members). They must consider weather conditions, travel time, location uncertainty, and rival boats when determining what decision to make. Our scenario was designed to be relatable to naive, civilian participants. Vignettes depicting scenes requiring complex decision making were created, and took the form of screenshots from the simulation testbed, which was developed in the Unity 3D game engine with an open source GIS plugin (see Fig. 1).

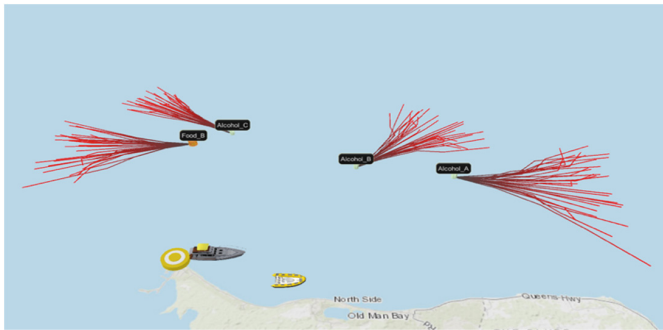


Fig. 1. Example a vignette extracted from the application illustrating the scenario.

2 Method

2.1 Experimental Design

We used a between-subjects design to evaluate the effects of training on decision making under uncertainty in a COA selection task.

Independent Variable. *Type of training* served as the independent variable and was manipulated at three levels: Control, Traditional, and Full.

Control Training. Control Training was the least informative condition. Participants read a brief introductory paragraph (see Fig. 2) before proceeding to the experimental tasks. We used this format based on previous studies with similar formats [11–13, 17].

Traditional Training. Traditional Training was designed to mimic what is sometimes found in the literature on visualization to support decision making [10]. Participants were presented with a text-only description of game components (see Fig. 3, left), along with rudimentary information about the game’s rules and scoring system.

On the following pages will be a series of test questions related to a task having to do with decision making and uncertainty. These include questions about different kinds of boats, party supplies, their location in the ocean, and weather. We are testing how well people can interpret questions about uncertainty and these kinds of things when they have been given no background information.

Following the test questions, you will be presented with a series of scenarios called The Party Game. You will have the opportunity to earn points in the game scenarios.

In addition to the base pay of \$2.00, if you score in the top 10%, you will receive a bonus of \$1.00.

Fig. 2. Control training introduction page.

Full Training. Based on prior work on computer-based training for complex tasks [19], we designed a more comprehensive training to help participants make inferences about and comparisons across task elements. The aim was to more thoroughly ground participants in the task environment to aid learning [20]. Full Training included the same information from Traditional Training, with the addition of images, tables, comparison charts, and supplemental text (see Fig. 3, right).

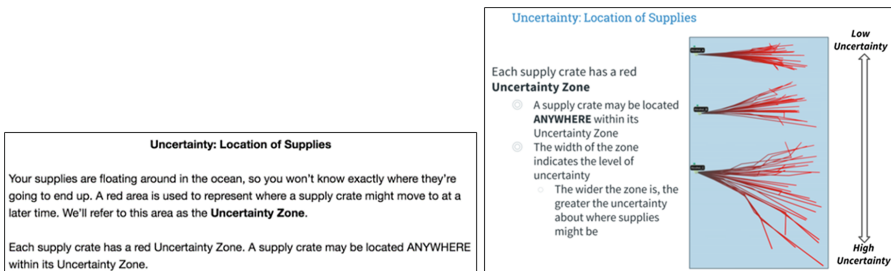


Fig. 3. Examples of equivalent information presented in Traditional Training (*left*) compared to Full Training (*right*).

In all conditions, training was self-paced with no time limit. Traditional and Full Training were split into multiple sections describing game elements, rules, and uncertainty visualization.

Dependent Variables. We investigated the effect of training on performance accuracy, workload, and cognitive efficiency. To obtain these measures, we developed assessments tapping both knowledge acquisition and knowledge application based on prior work on decision making in complex contexts [19, 20]. First, we had two measures of *knowledge acquisition*—the degree to which participants could recognize and understand basic concepts related to the decision-making task. First, the Recognition Knowledge Test assessed participants' recognition and identification of core training concepts (e.g., resources, supplies). Second, the Declarative Knowledge Test assessed learners' mastery of basic factual information associated with the training (e.g., value of assets). These tests each comprised ten multiple-choice questions with four response options, one of which was correct.

Next, we developed the Simulation Vignette Test to measure *knowledge application*, a complex form of assessment designed to tap a higher level of understanding by requiring integration of a variety of cues in a complex setting. Because uncertainty visualizations often support how to optimize COA selections, simulation vignettes were designed to mimic this while still being answerable by a naïve population. Participants were shown vignettes, consisting of images of a map with items (e.g., supplies, boats, etc.) with location uncertainty illustrated by spaghetti plots. The assessment consisted of ten multiple-choice questions. Participants were informed that for each vignette, there may be more than one correct response; however, they were instructed to select the *optimal* solution that maximized points. Each question had four response options: optimal, suboptimal (better), suboptimal (worse), and completely incorrect. For each vignette, all participants were provided with a “cheat sheet” with the information required to make a decision (e.g., capabilities of resources, value of supplies). We included this information so differences in performance could be credited to an understanding of how to apply knowledge acquired from training rather than memorization. Additionally, because the Control Training condition provided minimal information, this cheat sheet was necessary to allow all three conditions to complete the task. From these assessments, we obtained the following measures.

Performance Accuracy. We calculated the percent of correct responses for each test section (Recognition, Declarative, and Simulation Vignette). For the knowledge application (Simulation Vignette) assessment, performance accuracy was calculated as the percent of optimal uncertainty judgements.

Workload. Each question was accompanied by a subjective workload assessment asking participants to rate how easy or difficult it was to answer each item using a 7-point Likert-type scale. We calculated mean workload scores for each test section.

Cognitive Efficiency. We used a unique assessment of workload relative to performance. We drew from Cognitive Load Theory (CLT) and gains in understanding how human information processing influences decision making [21]. CLT attends to how a domain’s complexity can alter workload; for example, dealing with multiple resources may overwhelm the human information processing system. We also leveraged the concept of *cognitive efficiency*, a metric evolving out of instructional efficiency measures that describes the relationship between workload and performance [22–24]. A cognitive efficiency score is obtained by computing the difference between standardized performance accuracy and workload scores. Positive scores indicate higher relative performance compared to relative workload and suggest more efficient cognitive processing. Negative scores indicate lower relative performance compared to relative workload.

2.2 Research Hypotheses

Hypothesis 1: Performance Accuracy. The Full Training condition will outperform Traditional Training and Control Training. Specifically:

- 1a. Training condition will have a significant effect on a naive population's overall knowledge assessment accuracy.
- 1b. Training condition will have a significant effect on knowledge acquisition for the Recognition Knowledge Test (REC) with Full Training outperforming Traditional Training and Control Training.
- 1c. Training condition will have a significant effect on knowledge acquisition for the Declarative Knowledge Test (DEC) with Full Training outperforming Traditional Training and Control Training.
- 1d. Training condition will have a significant effect on knowledge application in the Simulation Vignette Test (VIG) with Full Training outperforming Traditional Training and Control Training.

Hypothesis 2: Workload. The Full Training condition will lead to lower subjective workload during testing compared to Traditional Training and Control Training. Specifically:

- 2a. Training condition will have a significant effect on a naive population's overall workload experienced during knowledge assessment.
- 2b. Training condition will have a significant effect on workload during knowledge acquisition for the Recognition Knowledge Test (REC) with Full Training leading to lower workload compared to Traditional Training and Control Training.
- 2c. Training condition will have a significant effect on workload during knowledge acquisition for the Declarative Knowledge Test (DEC) with Full Training leading to lower workload compared to Traditional Training and Control Training.
- 2d. Training condition will have a significant effect on workload during knowledge application for the Simulation Vignette Test (VIG) with Full Training leading to lower workload compared to Traditional Training and Control Training.

Hypothesis 3: Cognitive Efficiency. The Full Training condition will lead to higher cognitive efficiency during testing compared to Traditional Training and Control Training. Specifically:

- 3a. Training condition will have a significant effect on a naive population's overall cognitive efficiency during knowledge assessment.
- 3b. Training condition will have a significant effect on cognitive efficiency during knowledge acquisition for the Recognition Knowledge Test (REC) with Full Training leading to greater cognitive efficiency compared to Traditional Training and Control Training.
- 3c. Training condition will have a significant effect on cognitive efficiency during knowledge acquisition for the Declarative Knowledge Test (DEC) with Full Training leading to greater cognitive efficiency compared to Traditional Training and Control Training.

- 3d. Training condition will have a significant effect on cognitive efficiency during knowledge application for the Simulation Vignette Test (VIG) with Full Training leading to greater cognitive efficiency compared to Traditional Training and Control Training.

2.3 Participants

We recruited 247 participants (132 female), ranging in age from 21 to 70 (mean age = 35.68 years) from Amazon's Mechanical Turk (AMT). 87.45% of participants were located in the United States. Participants were eligible if they identified English as their primary language. Also, because some task elements had alcohol related themes, participants had to be at least 21 years old. Participants were compensated \$2.00 USD for completing the study. To increase effort and motivation, we awarded a \$1.00 bonus to participants scoring in the top 10% on the COA task.

2.4 Procedure

First, participants were directed to our survey hosted on Qualtrics. Following confirmation of age and informed consent, participants were randomly assigned to Control, Traditional, or Full Training. Following training (or introductory paragraph), all participants completed the knowledge acquisition assessment, consisting of the Recognition Test and Declarative Knowledge Test. Prior to both tests, participants completed a practice question and were given feedback about the correct response. Following the knowledge acquisition assessment, participants completed a practice task to familiarize them with the knowledge application assessment, and were given feedback about the correct solution. Participants then completed the Simulation Vignette Test. For all assessment sections, question order was randomized, as was order of response options. Finally, participants completed a demographic questionnaire. To determine rankings for bonuses, partial points were awarded. For each vignette, optimal (best) solutions counted as 1 point, suboptimal solutions counted as 0.5 (better) and 0.25 (worse) points, and the completely incorrect option counted as 0 points. Participants who scored in the top 10% based on these criteria were awarded a \$1.00 bonus.

3 Results

3.1 Performance Accuracy

To test Hypothesis 1, a MANOVA compared the effects of training type on accuracy for the three assessment types. There was a significant effect of training type on our three test types: Recognition Knowledge Test (REC), Declarative Knowledge Test (DEC), and Simulation Vignette Test (VIG). In support of Hypothesis 1a, training condition had a significant effect on overall performance accuracy (see Fig. 4), Wilk's Lambda $F(6, 484) = 8.81, p < .001$, observed power = 1.0. Accuracy was highest in Full Training ($M = .656, SD = .193$), next highest in Traditional Training ($M = .617, SD = .187$), and lowest in Control Training ($M = .514, SD = .170$).

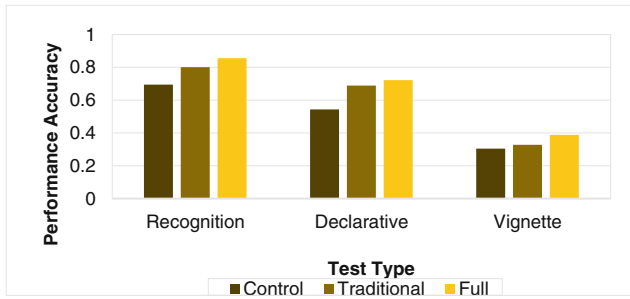


Fig. 4. Performance accuracy by condition and test type.

In support of Hypothesis 1b, training had a significant effect on REC performance accuracy, $F(2, 244) = 17.16$, $p < .001$, observed power = 1.00. REC accuracy was highest in Full Training ($M = .856$, $SD = .185$), next highest in Traditional Training ($M = .801$, $SD = .184$), and lowest in Control Training ($M = .695$, $SD = .162$). Post-hoc comparison showed accuracy for each condition was significantly different from one another.

In support of Hypothesis 1c, training had a significant effect on DEC performance accuracy, $F(2, 244) = 20.08$, $p < .001$, observed power = 1.00. DEC accuracy was highest in Full Training ($M = .723$, $SD = .213$), next highest in Traditional Training ($M = .686$, $SD = .191$) and lowest in Control Training ($M = .543$, $SD = .169$). Post-hoc comparison showed significant accuracy differences between Control Training and the others, but not between Traditional and Full Training.

In support of Hypothesis 1d, training had a significant effect on VIG performance accuracy, $F(2, 244) = 4.72$, $p < .01$, observed power = .786. VIG accuracy was highest in Full Training ($M = .389$, $SD = .181$), next highest in Traditional Training ($M = .328$, $SD = .177$), and lowest in Control Training ($M = .305$, $SD = .179$). Post-hoc comparison showed significant accuracy differences between Full Training and both other conditions, but not between Control and Traditional Training.

3.2 Workload

To test Hypothesis 2, a MANOVA compared the effects of training type on workload for the three assessment types. In support of Hypothesis 2a, training condition had a significant effect on overall workload (see Fig. 5), Wilk's Lambda $F(6, 484) = 12.49$, $p < .001$, observed power = 1.0. Workload was lowest in Full Training ($M = 3.05$, $SD = 1.35$), next lowest in Traditional Training ($M = 3.38$, $SD = 1.28$), and highest in Control Training ($M = 4.27$, $SD = 1.23$).

In support of Hypothesis 2b, training had a significant effect on REC workload, $F(2, 244) = 32.56$, $p < .001$, observed power = 1.00. REC workload was lowest in Full Training ($M = 2.23$, $SD = 1.32$), next lowest in Traditional Training ($M = 2.92$, $SD = 1.22$), and highest in Control Training ($M = 3.77$, $SD = 1.09$). Post-hoc comparison showed workload for each condition was significantly different from one another.

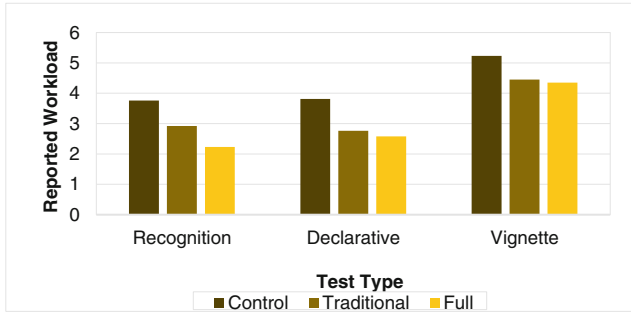


Fig. 5. Reported workload by condition and test type.

In support of Hypothesis 2c, training had a significant effect on DEC workload, $F(2, 244) = 23.15$, $p < .001$, observed power = 1.00. DEC workload was lowest in Full Training ($M = 2.58$, $SD = 1.25$), next lowest in Traditional Training ($M = 2.76$, $SD = 1.64$) and highest in Control Training ($M = 3.81$, $SD = 1.29$). Post-hoc comparison showed significant workload differences between Control Training and the other conditions, but not between Traditional and Full Training.

In support of Hypothesis 2d, training had a significant effect on VIG workload, $F(2, 244) = 9.53$, $p < .01$, observed power = .98. VIG workload was lowest in Full Training ($M = 4.35$, $SD = 1.48$), next lowest in Traditional Training ($M = 4.45$, $SD = 1.45$), and highest in Control Training ($M = 5.23$, $SD = 1.30$). Post-hoc comparison showed significant workload differences between Control Training and the other conditions, but not between Traditional and Full Training.

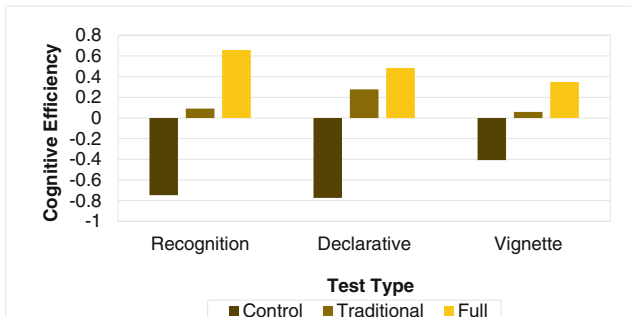


Fig. 6. Cognitive efficiency by condition and test type.

3.3 Cognitive Efficiency

In support of Hypothesis 3a, training condition had a significant effect on overall cognitive efficiency (see Fig. 6), Wilk's Lambda $F(6, 484) = 13.08$, $p < .001$, observed power = 1.0. Cognitive efficiency was highest in Full Training ($M = .497$, $SD = 1.2$), next highest in Traditional Training ($M = .142$, $SD = 1.05$), and lowest in Control Training ($M = -.642$, $SD = .97$).

In support of Hypothesis 3b, training had a significant effect on REC cognitive efficiency, $F(2, 244) = 33.74$, $p < .001$, observed power = 1.00. REC cognitive efficiency was highest in Full Training ($M = .658$, $SD = 1.25$), next highest in Traditional Training ($M = .091$, $SD = 1.11$), and lowest in Control Training ($M = -.746$, $SD = .890$). Post-hoc comparison showed cognitive efficiency for each condition was significantly different from one another.

In support of Hypothesis 3c, training had a significant effect on DEC cognitive efficiency, $F(2, 244) = 31.09$, $p < .001$, observed power = 1.00. DEC cognitive efficiency was highest in Full Training ($M = .485$, $SD = 1.20$), next highest in the Traditional Training ($M = .277$, $SD = 1.06$) and lowest in Control Training ($M = -.773$, $SD = 1.01$). Post-hoc comparison showed significant cognitive efficiency differences between Control Training and the others, but not between Traditional and Full Training.

In support of Hypothesis 3d, there was a significant effect of training on VIG cognitive efficiency, $F(2, 244) = 10.55$, $p < .001$, observed power = .99. Cognitive efficiency was highest on VIG in the Full Training condition ($M = .349$, $SD = 1.15$), next highest in Traditional Training ($M = .059$, $SD = .983$), and lowest in Control Training ($M = -.407$, $SD = 1.03$). Post-hoc comparison showed significant cognitive efficiency differences between Control Training and the other conditions, but not between Traditional and Full Training.

4 Discussion

Our overarching goal was to document the importance of training to support task comprehension and the ability to appropriately apply knowledge when making decisions under uncertainty. As predicted, the results document how appropriately designed training can facilitate both knowledge acquisition and knowledge application and thus contribute to improved evaluations of uncertainty visualization interventions.

Broadly, our results suggest that providing any training prior to a decision-making task involving uncertainty visualization improves overall performance. However, the effect of training on performance may vary for different tasks, and performance on certain tasks may benefit more from comprehensive training. Our results suggest traditional, text-only training may be sufficient to facilitate acquisition of factual knowledge when recognition of visual stimuli is not needed. However, when recognition and application of visual stimuli are required, traditional training may be insufficient. As such, to truly determine the effectiveness of visualizations, more comprehensive training may be needed for the complex tasks used in uncertainty visualization studies. Additionally, comprehensive training may aid in the reduction of cognitive effort during complex tasks compared to traditional training, and lack of adequate training can result in reduced information processing capacity, and thus, impaired performance. Importantly, our measure of cognitive efficiency provides an additional diagnostic for evaluating performance. As hypothesized, providing any training led to greater cognitive efficiency. Our results suggest basic training may sufficiently improve cognitive efficiency during tests of factual knowledge and decision-making tasks. However, comprehensive training was more beneficial for cognitive efficiency when identifying

visual stimuli, and overall performance on decision-making tasks benefits more from comprehensive training.

With respect to potential limitations, because the study was run online, we could not monitor participants or control for external distractions. However, one of the advantages of an online study was that we were not limited to a narrow segment of the population and tested a wider demographic. Furthermore, the use of a scenario that is more relatable to a civilian and naive population, in lieu of a traditional military scenario, bolsters the ecological validity of this study. Essentially, this study serves as a jumping-off point for improving uncertainty visualization research by providing evidence of the impact of training in studies with non-expert participants. Despite a general understanding of the importance of training, there is not a standard, empirically-supported approach to training for experimental tasks in this area. This highlights the need for research that focuses on the impact of training on performance in decision-making tasks when evaluating uncertainty visualizations.

5 Conclusion

In sum, we submit that without adequately introducing participants to experimental tasks, it is unclear how uncertainty visualization interventions influence performance. The present research not only suggests that training has a significant effect on how participants perform on a task involving uncertainty visualization, as well as on task comprehension, but our research also provides evidence that training is a valuable tool for improving the assessment of uncertainty visualization.

Acknowledgments. This work was supported by the Office of Naval Research Grant N00014-15-1-2708, under the Command Decision Making program. The views and opinions contained in this article are the authors' and should not be construed as official or as reflecting the views of the University of Central Florida or the Office of Naval Research.

References

1. Andrienko, G., Andrienko, N., Demsar, U., Dransch, D., Dykes, J., Fabrikant, S.I., Jern, M., Kraak, M.J., Schumann, H., Tominski, C.: Space, time and visual analytics. *Int. J. Geogr. Inf. Sci.* **24**, 1577–1600 (2010)
2. Sedig, K., Parsons, P.: Interaction design for complex cognitive activities with visual representations: a pattern-based approach. *AIS Trans. Hum. Comput. Interact.* **2**, 84–133 (2013)
3. Kehrler, J., Hauser, H.: Visualization and visual analysis of multifaceted scientific data: a survey. *IEEE Trans. Vis. Comput. Graph.* **19**, 495–513 (2013)
4. Sanyal, J., Zhang, S., Bhattacharya, G., Amburn, P., Moorhead, R.J.: A user study to compare four uncertainty visualization methods for 1D and 2D datasets. *IEEE Trans. Vis. Comput. Graph.* **15**, 1209–1218 (2009)
5. Bisantz, A.M., Cao, D., Jenkins, M., Pennathur, P.R., Farry, M., Roth, E., Potter, S.S., Pfautz, J.: Comparing uncertainty visualizations for a dynamic decision-making task. *J. Cogn. Eng. Decis. Mak.* **5**, 277–293 (2011)

6. MacEachren, A.M., Robinson, A., Hopper, S., Gardner, S., Murray, R., Gahegan, M., Hetzler, E.: Visualizing geospatial information uncertainty: what we know and what we need to know. *Cartogr. Geogr. Inf. Sci.* **32**, 139–160 (2005)
7. Nadav-Greenberg, L., Joslyn, S.L.: Uncertainty forecasts improve decision making among nonexperts. *J. Cogn. Eng. Decis. Mak.* **3**, 209–227 (2009)
8. Isenberg, T., Isenberg, P., Chen, J., Sedlmair, M., Moller, T.: A systematic review on the practice of evaluating visualization. *IEEE Trans. Vis. Comput. Graph.* **19**, 2818–2827 (2013)
9. Andrews, K.: Evaluation comes in many guises. In: *AVI Workshop on BEyond Time Errors*, pp. 8–10 (2008)
10. Kwon, B.C., Lee, B.: A comparative evaluation on online learning approaches using parallel coordinate visualization. In: *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, pp. 993–997 (2016)
11. Netzel, R., Hlawatsch, M., Burch, M., Balakrishnan, S., Schmauder, H., Weiskopf, D.: An evaluation of visual search support in maps. *IEEE Trans. Vis. Comput. Graph.* **23**, 421–430 (2017)
12. Chang, C., Bach, B., Marriott, K., Dwyer, T.: Evaluating perceptually complementary views for network exploration tasks. In: *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, pp. 1397–1407 (2017)
13. Leitner, M., Battenfield, B.P.: Guidelines for the display of attribute certainty. *Cartogr. Geogr. Inf. Sci.* **27**, 3–14 (2000)
14. Moray, N.: Subjective mental workload. *Hum. Factors J. Hum. Factors Ergon. Soc.* **24**, 25–40 (1982)
15. Kirschenbaum, S.S., Trafton, J.G., Schunn, C.D., Trickett, S.B.: Visualizing uncertainty: the impact on performance. *Hum. Factors* **56**, 509–520 (2014)
16. Karstens, C.D., Stumpf, G., Ling, C., Hua, L., Kingfield, D., Smith, T.M., Correia, J., Calhoun, K., Ortega, K., Melick, C., Rothfusz, L.P.: Evaluation of a probabilistic forecasting methodology for severe convective weather in the 2014 hazardous weather testbed. *Weather Forecast.* **30**, 1551–1570 (2015)
17. Loft, S., Morrell, D.B., Ponton, K., Braithwaite, J., Bowden, V., Huf, S.: The impact of uncertain contact location on situation awareness and performance in simulated submarine track management. *Hum. Factors* **58**, 1052–1068 (2016)
18. Beach, L.R., Mitchell, T.R.: A contingency model for the selection of decision strategies. *Acad. Manag. Rev.* **3**, 439–449 (1978)
19. Cuevas, H.M., Fiore, S.M., Bowers, C.A., Salas, E.: Fostering constructive cognitive and metacognitive activity in computer-based complex task training environments. *Comput. Hum. Behav.* **20**, 225–241 (2004)
20. Fiore, S.M., Cuevas, H.M., Scielzo, S., Salas, E.: Training individuals for distributed teams: problem solving assessment for distributed mission research. *Comput. Hum. Behav.* **18**, 729–744 (2002)
21. Paas, F., Renkl, A., Sweller, J.: Cognitive load theory and instructional design: recent developments. *Educ. Psychol.* **38**, 1–4 (2003)
22. Fiore, S.M., Scielzo, S., Jentsch, F., Howard, M.L.: Effects of discrimination task training on X-ray screening decisions. In: *Proceedings of the 50th Annual Meeting of the Human Factors and Ergonomics Society*, pp. 2610–2614 (2006)
23. Paas, F.G.W.C., Van Merriënboer, J.J.G.: The efficiency of instructional conditions: an approach to combine mental effort and performance measures. *Hum. Factors J. Hum. Factors Ergon. Soc.* **35**, 737–743 (1993)
24. Johnston, J.H., Fiore, S.M., Paris, C., Smith, C.A.P.: Application of cognitive load theory to develop a measure of team cognitive efficiency. *Mil. Psychol.* **25**, 252–265 (2013)